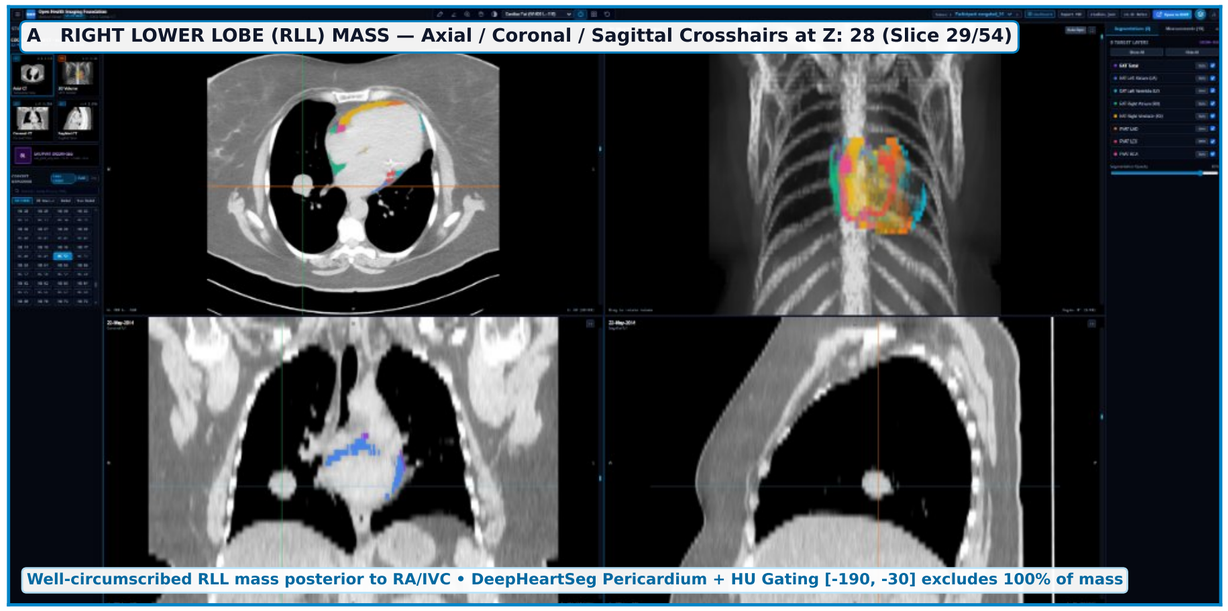
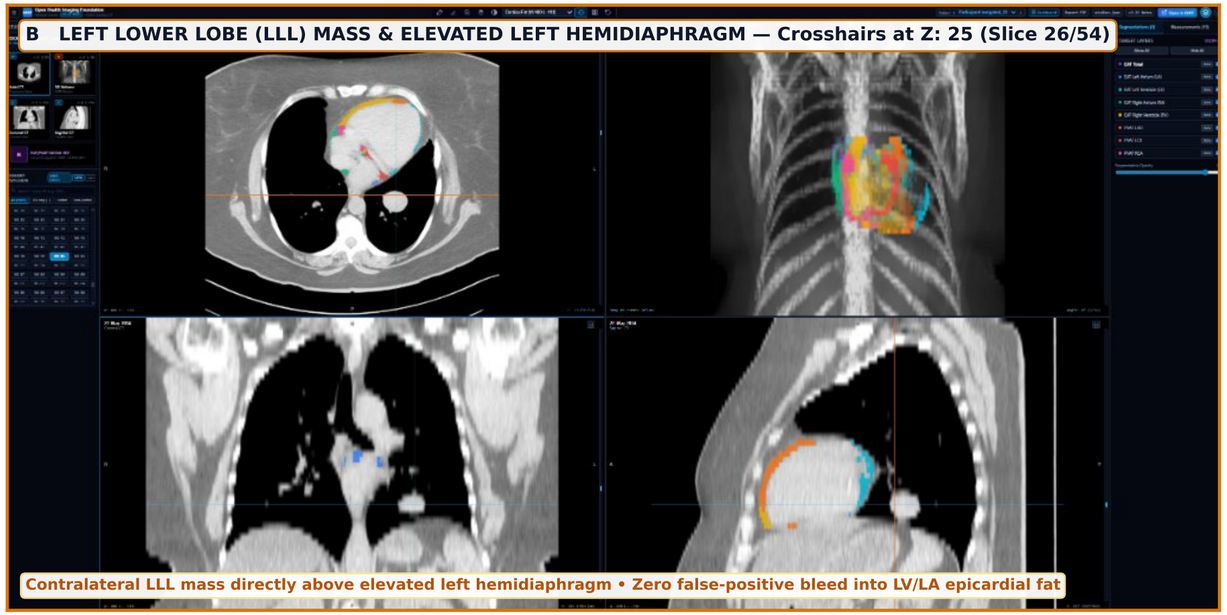
STANFORD COCA CASE STUDY — PARTICIPANT nongated_51: 3D FAT QUANTIFICATION & VLM DISCOVERY

EAT: 118.4 cm³ • LCX PVAT: -60.6 HU (INFLAMED) • RCA PVAT: -63.6 HU (INFLAMED)

8-Compartment Epicardial/Pericoronary Fat Segmentation (Zero Bleed into Paracardiac Masses) + NV-Reason-CT Bilateral Lower Lobe Mass Detection

NV-Reason-CT (3D ViT + Qwen3.5-4B): Bilateral Lower Lobe Masses + Left Hemidiaphragm Elevation





C 8-COMPARTMENT CARDIAC ADIPOSE & CRISP-CT INFLAMMATION QUANTIFICATION

Compartment / Biomarker	Volume (cm³)	Mean HU	Clinical Interpretation / Threshold
Total Epicardial Fat (EAT)	**118.4 cm³**	**-67.6 HU**	**Moderate Burden (~68th %ile; Normal < 100 cm³)**
• Right Ventricle (EAT RV)	63.6 cm³	-67.2 HU	Dominant anterior/inferior RV fat pad
• Left Ventricle (EAT LV)	38.4 cm³	-68.8 HU	Intact posterior LV boundary (0% LLL mass bleed)
• Left Atrium (EAT LA)	26.2 cm³	-66.4 HU	Periatrial adipose envelope
• Right Atrium (EAT RA)	26.1 cm³	-67.9 HU	Intact posterior RA boundary (0% RLL mass bleed)
Total Pericoronary Fat (PVAT)	**30.5 cm³**	**-66.5 HU**	**10mm 3D Anisotropic EDT Perivascular Cuff**
• LAD Pericoronary (PVAT LAD)	13.2 cm³	-71.8 HU	NORMAL (<= -70.1 HU CRISP-CT threshold)
• LCX Pericoronary (PVAT LCX)	**7.1 cm³**	**-60.6 HU**	**ACTIVE INFLAMMATION (> -70.1 HU; +9.5 HU shift)**
• RCA Pericoronary (PVAT RCA)	**10.1 cm³**	**-63.6 HU**	**ACTIVE INFLAMMATION (> -70.1 HU; +6.5 HU shift)**

D NVIDIA NV-REASON-CT (3D ViT + Qwen3.5-4B) — VERBATIM AI RADIOLOGIST REPORT

TECHNIQUE:	Non-contrast CT (COCA Non-Gated Chest/Cardiac CT • 54 slices • 3.0 mm).
Lungs and airways:	**Bilateral lower lobe lung masses. Left hemidiaphragm elevation. Central airways are patent.**
Pleura / Mediastinum:	No pleural effusion or pneumothorax. No hilar lymphadenopathy or mediastinal mass.
Heart & Pericardium:	Normal heart size. No pericardial effusion. Normal caliber thoracic aorta and main PA.
IMPRESSION:	**1. Bilateral lower lobe lung masses. 2. Left hemidiaphragm elevation.**
KEY FINDINGS TAG:	**Lung mass**

WHY COMPLEMENTARY 3D SEGMENTATION + 3D VLM REASONING MATTERS IN CLINICAL PRACTICE:

1. Orthogonal Physics & Anatomical Gating: DeepHeartSeg ONNX pericardial mesh + adipose HU thresholding [-190, -30 HU] prevents soft-tissue tumors (+20 to +50 HU) abutting the heart from corrupting EAT/PVAT volumes, while detecting severe LCX (-60.6 HU) & RCA (-63.6 HU) inflammation.
2. Zero-Shot Whole-Thorax Safety Net: Standard cardiac calcium pipelines ignore extra-cardiac lung parenchyma; NV-Reason-CT inspects the full 192³ LPS volume and accurately flags both the Right Lower Lobe mass (Z=28), Left Lower Lobe mass (Z=25), and Left Hemidiaphragm Elevation.